

Jeonghan Yu

Doctor of Philosophy in Mechanical Engineering

239, S.W. Mudd, Broadway & 120th street, New York, NY 10027, USA

☎ (+1) 551-373-0010, (+82) 10-6247-9948 | ✉ jy3471@columbia.edu, sailoryu@snu.ac.kr | 🌐 jeonghan-yu

Research Interests

“Design, Control, Experimentation, and Biomechanical Analysis of a Wearable Robotic/Rehabilitation System”

Wearable Robotics/Rehabilitation Robotics

- Hardware and software development of rehabilitation robotics;
- Rigid-body mechanism/cable-based robots for individuals with cerebral palsy and spinal cord injury;
- Applications (so far): exoskeleton (self-aligning knee, moment-varying hip, wheelchair-mounted), cable-driven robots (stand and sitting trainers)
- AI-driven assistance: Autonomous pressure relief under buttocks for wheelchair users; Human-robot interaction)
- Control: Adaptive impedance controller based on Gaussian mixture model; Motion predictive controller using ground reaction force; Feedforward model for flexible shaft-driven exoskeleton; Real-time and FPGA motor control
- Biomechanical analysis using motion capture, EMG, and ground reaction forces

Autonomous Design Algorithm of Linkage Mechanisms

- Topology optimization of linkage mechanisms; Gradient-based optimization; Sensitivity analysis;
- Formulation development considering robot kinematics (DOFs, mechanical advantage, displacement constraints);
- Nonlinear optimization for inverse problems; Mechanism ground models

Education

Seoul National University

Seoul, S.Korea

Ph.D., Department of Mechanical Engineering

09/01/2017 - 02/26/2024

- Thesis: **Topology Optimization of Self-Aligning Wearable Exoskeleton Robot Mechanism**
- Advisor: Dr. Yoon Young Kim (Distinguished Professor)

Seoul National University

Seoul, S.Korea

B.S., Department of Mechanical and Aerospace Engineering

03/01/2013 - 02/24/2017

- Thesis: Simultaneous Shape and Topology Optimization of Planar Linkage Mechanisms Based on the Spring-connected Rigid Block Model

Research Experiences

Department of Mechanical Engineering, Columbia University

New York, USA

Postdoctoral Research Scientist

04/2024 - Current

- Principal Investigator: Prof. Sunil Agrawal (Columbia University)
- Topic: Biomechanical analysis on whole-body movements, Training of spinal cord-injured patients to improve or maintain their functional activity such as standing or reaching with their hands while sitting; all aspects of innovative rehabilitation of these patients, including robot system design, programming, testing, data collection, and data analysis.

Soft Robotics Research Center, Seoul National University

Seoul, S.Korea

Postdoctoral Researcher

03/2024

- Principal Investigator: Prof. Frank Chongwoo Park (Seoul National University)
- Topic: Optimization of robot design and control

Chung-Ang University

Seoul, S.Korea

Postdoctoral Researcher

01/2024 - 04/2024

- Principal investigator: Prof. Giuk Lee (Chung-Ang University)

- Topic: Wearable exoskeleton robot design, control, and experimentation, hip balance assistance strategy

Seoul National University

Seoul, S.Korea

Laboratory Student Representative (Elected/Volunteered)

07/2021 - 06/2022

- Leading 35 researchers (Post-docs., Ph.D., and MS students) among Prof. Kim's group and communicating with Prof. Kim and researchers

- Advisor: Prof. Yoon Young Kim (Seoul National University)
- Topic: Topology optimization design of linkage mechanisms, Autonomous creation of artworks (Variational art)

Awards

- 04/2024 **Best Paper Award**
2023 KSME Dynamics and Control Division, The Korean Society of Mechanical Engineers (KSME)
Winning Paper: Flexible Shaft Control for Power Transmission of Wearable Exoskeleton Robots
- 05/2022 **Best Student Presentation Award**
2021 KSME CAE and Applied Mechanics Division, The Korean Society of Mechanical Engineers (KSME)
Winning Paper: Topology Optimization Design of an Exoskeleton Self-Alignment Mechanism Joint Considering Mechanical Advantage
- 02/2022 **The 7th Place Winner in Mechanical Engineering (Total: 2,105 Participants), Participation Prize**
The 28th Humantech Paper Award, Samsung
Winning paper: Autonomous Physics-Driven Design of Exo-Robot Knee Joint Mechanisms: A New Paradigm
- 05/2017 **The 4th Honorable Mention**
2016 SNU Undergraduate Research, Seoul National University
Winning Project: Design of a New Concept Swimming Mechanism Based on Linkage Mechanism
- 12/2016 **Best Presentation Award**
Department of Mechanical and Aerospace Engineering, Seoul National University
Winning Paper: Simultaneous Shape and Topology Optimization of Planar Linkage Mechanisms Based on the Spring-connected Rigid Block Model

Publications

- [1] Jeonghan Yu, Isirame Omofuma, Darren Biskup, Himanshu Pota, Sunil K. Agrawal*, “Whole-Body Coordination and Spinopelvic Alignment during Yoga Standing Maneuvers: Comparing Practitioners with Non-Practitioners,” *IEEE Transactions on Neural Systems and Rehabilitation Engineering* (JIF: 6.6(Q1), JCI: 2.02(Q1)), Vol. 34, pp. 2773-2785, May 2026, DOI: 10.1109/TNSRE.2026.3697342
- [2] Jegyeong Ryu, Jongjun Lee, Jeonghan Yu, Seok Won Kang*, and Yoon Young Kim*, “Virtual Framework Customizing Wearable Mechanisms: An Application to Gait-Assistive Mobile Hip Robots,” *Robotics and Autonomous Systems* (JIF: 6.0(Q1), JCI: 1.01(Q1)), Vol. 192, p. 105052, October 2025, DOI: 10.1016/j.robot.2025.105052
- [3] Jeonghan Yu, Seok Won Kang*, and Yoon Young Kim*, “Autonomous Synthesis of Self-Aligning Knee Joint Exoskeleton Mechanisms,” *IEEE Transactions on Robotics* (JIF: 11.1(Q1), JCI: 2.32(Q1)), Vol. 41, pp. 2358-2373, March 2025, DOI: 10.1109/TRO.2025.3547274
- [4] Jongjun Lee, Jegyeong Ryu, Gyuik Lee, Jeonghan Yu, Seok Won Kang*, and Yoon Young Kim*, “Single-Actuator Hip Exoskeleton Mechanism Design with Actuation in Frontal and Sagittal Planes: Design Methodology and Experimental Validation,” *IEEE Access* (JIF: 4.2(Q2), JCI: 0.82(Q2)), Vol. 11, pp. 136630-136642, November 2023, DOI: 10.1109/ACCESS.2023.3337144
- [5] Jeonghan Yu, Sang Min Han, and Yoon Young Kim*, “Simultaneous Shape and Topology Optimization of Planar Linkage Mechanisms Based on the Spring-Connected Rigid Block Model,” *Journal of Mechanical Design* (JIF: 3.3(Q2), JCI: 0.80(Q1)), Vol. 142, No. 1, pp. 011404, January 2020, DOI: 10.1115/1.4044327

Under Review

- [*a] Jeonghan Yu, Sunil K. Agrawal*, “Proficiency-Dependent Postural Stability and Force Organization during Symmetric and Asymmetric Yoga Maneuvers while Standing,” *IEEE Transactions on Neural Systems and Rehabilitation Engineering* (JIF: 6.6(Q1), JCI: 2.02(Q1)), 2026

- [*b] Byung Wook Kim, Edward Choi, Jaewon Cho, **Jeonghan Yu**, Chawin Ophaswongse, Sunil K. Agrawal*, “**PRISMNet: Cross-Subject Multimodal Seated Posture Classification**,” *IEEE Journal of Biomedical and Health Informatics (JIF: 7.7(Q1), JCI: 1.77(Q1))*, 2026
- [*c] Darren Biskup, **Jeonghan Yu**, Zibo Yu, Sunil K. Agrawal*, “**Ground Reaction Force-Aware Model Predictive Control for Underactuated Upright Physical Assistance**,” *IEEE Robotics and Automation Letters (JIF: 5.3(Q2), JCI: 1.17(Q1))*, 2026

Patents

- [1] **Jeonghan Yu**, Seok Won Kang, and Yoon Young Kim, “**Knee Joint Guide Apparatus**,” Korea: 10-2638771, February 15, 2024, US: 12383417, August 12, 2025, **Selected item of prototype supporting program of Seoul National University, Representative exhibition work for the SNU Inno Tech Day, SNU, Korea, Representative exhibition work for CES 2023, Las Vegas, USA**

Activities

CES 2023 Exhibition

Las Vegas, USA

- Product: Chameleon Knee Robot
- SNU Pavilion in Eureka Park

01/2023

The Rising Stars Women in Engineering Workshop, Asian Deans' Forum 2022

Australia

- Representative of SNU (1 of 4 SNU Ph. D.'s and Ph. D. Candidates)
- University of New South Wales

10/2022 - 11/2022

Conferences

International Conferences

- [1] (Oral, Poster) **Jeonghan Yu**, Jongjun Lee, Seungtae Yang, Junyoung Moon, Yoon Young Kim, and Giuk Lee, “**Control of a Flexible Shaft-Driven Hip Exoskeleton Mechanism Emulator for Gait Assistance**,” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Detroit, Michigan, US, October 1-5, 2023
- [2] (Oral) **Jeonghan Yu**, Seok Won Kang, and Yoon Young Kim, “**Mechanism Topology Optimization of Self-Aligning Knee Exoskeletons**,” *the 15th World Congress on Structural and Multidisciplinary Optimization (WCSMO)*, Cork, Ireland, June 5-9, 2023
- [3] (Oral) **Jeonghan Yu**, Seok Won Kang, and Yoon Young Kim, “**Noble Design of Adaptive and Efficient Knee Exoskeletons by Mechanism Topology Optimization Method**,” *the 15th World Congress on Computational Mechanics (WCCM-XV) and 8th Asian Pacific Congress on Computational Mechanics (APCOM-VII)*, Yokohama, Japan, July 31 – August 5, 2022
- [4] (Oral) Neung Hwan Yim, **Jeonghan Yu**, Jungcho Kim, Seok Won Kang, and Yoon Young Kim, “**Why Are Spring-connected Block Models Useful for Topology Optimization of Rigid-body Linkage Mechanisms?**,” *Asian Congress on Structural and Multidisciplinary Optimization (ACSMO) 2022*, Matsue, Japan, May 22-26, 2022
- [5] (Oral) **Jeonghan Yu**, Sang Min Han, and Yoon Young Kim, “**Path Generation of Planar Linkage Mechanisms by Simultaneous Shape and Topology Optimization Based on the Spring-connected Rigid Block Model**,” *the 13th World Congress on Structural and Multidisciplinary Optimization (WCSMO)*, Beijing, China, May 20-24, 2019
- [6] (Oral) **Jeonghan Yu**, Sang Min Han, and Yoon Young Kim, “**Simultaneous Shape and Topology Optimization of Planar Linkage Mechanisms Based on the Spring-connected Rigid Block Model**,” *Asian Congress on Structural and Multidisciplinary Optimization (ACSMO) 2018*, Dalian, China, May 21-24, 2018
- [7] (Oral) Seok Won Kang, **Jeonghan Yu**, Sang Min Han, and Yoon Young Kim, “**Topology Optimization of General-Joint Planar Linkage Mechanisms with an Application to Finger Rehabilitation Device Design**,” *the 12th World Congress on Structural and Multidisciplinary Optimization (WCSMO)*, Braunschweig, Germany, June 5-9, 2017

Conferences held in South Korea

- [1] (Oral) **Jeonghan Yu**, Jongjun Lee, Seungtae Yang, Junyoung Moon, Giuk Lee, and Yoon Young Kim, “**Flexible Shaft Control for Power Transmission of Wearable Exoskeleton Robots**,” *the Korean Society of Mechanical Engineers (KSME) 2023 Spring Annual Meeting – Dynamics and Control Division*, Busan, Korea, May 17-20, 2023 (**Best Paper Award**)
- [2] (Poster) Gyuik Lee, Jongjun Lee, **Jeonghan Yu**, Youngsoo Kim, Giuk Lee, and Yoon Young Kim, “**Performance Evaluation of Spherical Mechanism in Hip Joint Exoskeleton Robot for Centroidal Alignment**,” *Samsung Global Symposium*, Seoul, Korea, April 21, 2023
- [3] (Oral) **Jeonghan Yu**, Seok Won Kang, and Yoon Young Kim, “**Topology Optimization Design of an Exoskeleton Self-Alignment Mechanism Joint Considering Mechanical Advantage**,” *the Korean Society of Mechanical Engineers (KSME) 2021 Spring Annual Meeting - CAE and Applied Mechanics Division*, Online, August 25-26, 2021 (**Best Student Presentation Award**)
- [4] (Oral) Jegyeong Ryu, Seok Won Kang, Jongjun Lee, **Jeonghan Yu**, and Yoon Young Kim, “**Simulation-Based Shape Optimization of Wearable Hip Robots to Reduce Metabolic Energy Cost in Walking**,” *the Korean Society of Mechanical Engineers (KSME) 2021 Spring Annual Meeting - CAE and Applied Mechanics Division*, Online, August 25-26, 2021
- [5] (Oral) **Jeonghan Yu**, Sang Min Han, and Yoon Young Kim, “**Synthesis of Planar Linkage Mechanisms by Concurrent Shape and Topology Optimization Using SBM**,” *the Korean Society of Mechanical Engineers (KSME) 2018 Spring Annual Meeting - CAE and Applied Mechanics Division*, Yeosu, Korea, April 26-28, 2018
- [6] (Oral) Seok Won Kang, **Jeonghan Yu**, Sang Min Han, and Yoon Young Kim, “**Design of a Linkage Mechanism for Finger Rehabilitation Device by Using a New Topology Optimization Method**,” *the Korean Society of Mechanical Engineers (KSME) 2017 Spring Annual Meeting - CAE and Applied Mechanics Division*, Jeju, Korea, May 25-27, 2017
- [7] (Oral) Seok Won Kang, **Jeonghan Yu**, Sang Min Han, and Yoon Young Kim, “**Controlling Complicated Joints in Automatic Synthesis of Planar Linkage Mechanisms**,” *the Korean Society of Mechanical Engineers (KSME) 2016 Annual Meeting - CAE and Applied Mechanics Division*, Gangwon, Korea, December 14-16, 2016
- [8] (Poster) **Jeonghan Yu**, Sang Min Han, Seok Won Kang, and Yoon Young Kim, “**Simultaneous Shape and Topology Optimization of Planar Linkage Mechanisms Based on the Spring-connected Rigid Block Model**,” *the Korean Society of Mechanical Engineers (KSME) 2016 Spring Annual Meeting - CAE and Applied Mechanics Division*, Jeju, Korea, April 7-9, 2016

Skills

Robotics	Design, development, control, operation (with human subjects including patients) FPGA and Real-time controller design, Admittance control, Motor control
Optimization	Topology optimization, Shape optimization, Gradient-based optimization, Non-gradient-based optimization, Development of optimization functions based on mechanism kinematics
Kinematics	Computational kinematic analysis, Design of linkage mechanism
Biomechanics	Musculoskeletal analysis, Motion and gait analysis, Human subject experimentation
AI	Classification, Reinforcement learning for design and control
Manufacturing	Prototyping, 3D printing, Wearable exoskeleton system design
Software	LabVIEW, SOLIDWORKS, VICON Nexus, Unity, OpenSim, ADAMS, OptiTrack Motive
Language	MATLAB, Python, C#

Project Experiences

- 2024 – Current **Improving Balance after Spinal Cord Injury Using a Robotic Upright Stand Trainer**
 Researcher, Supported by Translational Research Projects (TRP) in New York State Spinal Cord Injury Research Board, New York State Department of Health
- 2024 **Wheelchair Robot for Active Postural Support (WRAPS) of People with Spinal Cord Injury**
 Researcher, Innovative, Developmental, or Exploratory Activities (IDEA) Grant in Spinal Cord Injury. Supported by New York State Spinal Cord Injury Research Board (SCIRB), New York State Department of Health
- 2019 – 2023 **Creative Topology Design Methodology for High-Agility and High-Adaptability Robot Mechanisms**
 Researcher, Supported by Samsung Science & Technology Foundation

- 2016 – 2019 **Development and Application of Creative Spatial Mechanism Synthesis Technology**
Researcher, Supported by the National Research Foundation of Korea
- 2018 **Development of Human Body Conformable Hip Joint Mechanism for Exoskeleton Robot**
Researcher, Supported by Hyundai NGV
- 2016 **Topology Optimization of New Concept Vehicle Suspension Mechanisms**
Researcher, Supported by Hyundai NGV
- 2016 **Design of a New Concept Swimming Mechanism Based on Linkage Mechanism**
Project Leader, Supported by Seoul National University
- 2015 **Engineering Technology-Based User Participation Automatic Design**
Researcher, Supported by the National Research Foundation of Korea

Seminars, Workshops, and Lectures

- 17/08/23 **Mechanism Topology Optimization of Knee Self-Aligning Exoskeleton**
International Structural Optimization Workshop, presented at Hanyang University, Korea
- 14/06/23,
03/04/23 **Engineering Design Case: From Problem Definition to Prototype Production**

Class: Capstone Design Track-Project, Adventure Design, in Dept. of Mechanical, Robotics, and Energy Engineering at Dongguk University, Korea
- 18/08/22
- 19/08/22 **Autonomous Design Technology of Mechanism Topology**

Korean Society of Mechanical Engineers (KSME) & Institute of Control, Robotics, and Systems (ICROS)

Teaching & Advising Experiences

Teaching Assistants

SNU

- Elasticity (M2794.007400), Spring 2023
- Solid Mechanics (M2794.001000.001), Spring 2021
- Optimal Design (M2794.003600.001), Spring 2018

Research Mentor

SNU & CU

- Development of real-time visual feedback GUI during training protocol
- Topology optimization methodology for synthesizing mechanisms with crossed links
- Physics-informed neural network-based mechanism synthesis
- Optimal design of knee exoskeleton joint for improving balancing during gait
- Optimal design based on the spring-connected rigid block model for pitching motion assistance exoskeletons
- Calculation of optimal hip-joint moment profile for gait assistance based on motion capture and simulation

Academic Service

Journal Reviewer

- ASME Journal of Mechanisms and Robotics
- ASME Journal of Biomechanical Engineering
- IEEE Transactions on Mechatronics, IEEE/RSJ International Conference on Intelligent Robots and Systems

Scholarship

- 2017 - 2019 **BK21+ Research Scholarship**
Brain Korea 21 Plus

- 2018 **Korea Rotary Scholarship**
The Rotary Club in Korea
- 2016, 2019 **Academic Honor Scholarship**
Seoul National University
- 2014 – 2016 **Human Resource Development Scholarship**
Hyungsong Educational and Cultural Foundation